

5 Option byte description

There are six option bytes. They are configured by the end user depending on the application requirements. As a configuration example, the watchdog may be selected in hardware or software mode.

A 32-bit word is split up as follows in the option bytes.

Table 12. Option byte format

31-24	23-16	15 -8	7-0
Complemented option byte1	Option byte 1	Complemented option byte0	Option byte 0

The organization of these bytes inside the information block is as shown in [Table 13](#).

The option bytes can be read from the memory locations listed in [Table 13](#) or from the Option byte register (FLASH_OBR).

Note: The new programmed option bytes (user, read/write protection) are loaded after a system reset.

Table 13. Option byte organization

Address	[31:24]	[23:16]	[15:8]	[7:0]
0x1FFF F800	nUSER	USER	nRDP	RDP
0x1FFF F804	nData1	Data1	nData0	Data0
0x1FFF F808	nWRP1	WRP1	nWRP0	WRP0
0x1FFF F80C	nWRP3 ⁽¹⁾	WRP3 ⁽¹⁾	nWRP2 ⁽¹⁾	WRP2 ⁽¹⁾

1. Available only on STM32F303xB/C and STM32F358xC devices. Must be kept at reset values in STM32F303x6/8 and STM32F328x8 devices.

Table 14. Description of the option bytes

Flash memory address	Option bytes
0x1FFF F800	Bits [31:24]: nUSER Bits [23:16]: USER: User option byte (stored in FLASH_OBR[15:8]) This byte is used to configure the following features: - Select the watchdog event: Hardware or software - Reset event when entering Stop mode - Reset event when entering Standby mode Bit 23: Reserved Bit 22: SRAM_PE The SRAM hardware parity check is disabled by default. This bit allows the user to enable the SRAM hardware parity check. 0: Parity check enabled. 1: Parity check disabled. Bit 21: VDDA_MONITOR This bit selects the analog monitoring on the VDDA power source: 0: VDDA power supply supervisor disabled. 1: VDDA power supply supervisor enabled. Bit 20: nBOOT1 Together with the BOOT0 pin, this bit selects Boot mode from the main Flash memory, SRAM or System memory. Refer to Section 3.2 on page 53. Bit 19: Reserved, must be kept at reset. Bit 18: nRST_STDBY 0: Reset generated when entering Standby mode. 1: No reset generated. Bit 17: nRST_STOP 0: Reset generated when entering Stop mode 1: No reset generated Bit 16: WDG_SW 0: Hardware watchdog 1: Software watchdog Bits [15:8]: nRDP Bits [7:0]: RDP: Read protection option byte The value of this byte defines the Flash memory protection level 0xAA: Level 0 0xFF (except 0xAA and 0xCC): Level 1 0xCC: Level 2 The protection levels are stored in the Flash_OBR Flash option bytes register (RDPRT bits).
0x1FFF F804	Datax: Two bytes for user data storage. These addresses can be programmed using the option byte programming procedure. Bits [31:24]: nData1 Bits [23:16]: Data1 (stored in FLASH_OBR[31:24]) Bits [15:8]: nData0 Bits [7:0]: Data0 (stored in FLASH_OBR[23:16])

Table 14. Description of the option bytes (continued)

Flash memory address	Option bytes
0x1FFF F808	WRP_x: Flash memory write protection option bytes Bits [31:24]: nWRP1 Bits [23:16]: WRP1 (stored in FLASH_WRP[15:8]) Bits [15:8]: nWRP0 Bits [7:0]: WRP0 (stored in FLASH_WRP[7:0]) 0: Write protection active 1: Write protection not active Refer to Section 4.3.2: Write protection for more details. <i>Note: Even if WRP2 and WRP3 are not available, they must be kept at reset value.</i>
0x1FFF F80C	WRP_x: Flash memory write protection option bytes (available only on STM32F303xB/C and STM32F358xC) Bits [31:24]: nWRP3 Bits [23:16]: WRP3 (stored in FLASH_WRP[31:24]) Bits [15:8]: nWRP2 Bits [7:0]: WRP2 (stored in FLASH_WRP[23:16]) One bit of the user option bytes WRP_x is used to protect 2 pages of 2 Kbytes in the main memory block. 0: Write protection active 1: Write protection not active In total, 4 user option bytes are used to protect the whole main Flash memory. WRP0: Write-protects pages 0 to 15 WRP1: Write-protects pages 16 to 31 WRP2: Write-protects pages 32 to 47 ⁽¹⁾ WRP3: bits 0-6 write-protect pages 48 to 61, bit 7 write-protects pages 62 to 127 on STM32F303xB/C and STM32F358xC, pages 62 to 255 on STM32F303xD/E.

1. Available only on STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices. Must be kept at reset values in STM32F303x6/8 and STM32F328x8 devices.

On every system reset, the option byte loader (OBL) reads the information block and stores the data into the Option byte register (FLASH_OBR) and the Write protection register (FLASH_WRP). Each option byte also has its complement in the information block. During option loading, by verifying the option bit and its complement, it is possible to check that the loading has correctly taken place. If this is not the case, an option byte error (OPTERR) is generated. When a comparison error occurs, the corresponding option byte is forced to 0xFF. The comparator is disabled when the option byte and its complement are both equal to 0xFF (Electrical Erase state).